

To comply with SEBI's regulations, businesses planning to migrate to the cloud need to select a provider with data centres located in India. They must also ensure that the provider has appropriate security measures in place to meet SEBI's encryption requirements.

Requirements/ Constraints	Example (Stock brokerage industry in India)
Compliance	Data storage and encryption regulations set forth by the Securities and Exchange Board of India (SEBI)
Budget	Limited budget for cloud services
Staffing	IT team with modest capabilities
Existing IT infrastructure	Legacy systems or applications that may not be compatible with cloud solutions

By using a simple table matrix like the one given above, businesses can identify their requirements and constraints, facilitating an informed decision-making process.

In addition to compliance requirements, businesses must also consider other constraints, such as budget, staffing, and existing IT infrastructure. For example, a business with limited budget may need to choose a cloud provider with more cost-effective pricing models, while a business with a small IT team may need to choose a cloud provider that offers more managed services to help with maintenance and support.

Technical factors to consider

When migrating workloads to the cloud, businesses need to evaluate their existing IT infrastructure to determine cloud readiness, choose the right deployment model and address all security concerns.

Evaluating existing IT infrastructure and determining cloud readiness: To determine cloud readiness, businesses should consider the following factors.

Application compatibility: Evaluate which applications can be moved to the cloud seamlessly and identify those requiring modifications to operate effectively in the cloud environment. AWS recommends employing migration strategies like 'Rehost' and 'Refactor' to address application compatibility concerns.

Resource utilisation: Assess the current resource usage including CPU, memory, storage, and network bandwidth, and verify that the cloud environment can accommodate the workload. The 'Rehost' strategy is well-suited for applications requiring minimal changes.

Data dependencies: Identify applications with critical data dependencies and analyse the impact of migrating them to the

cloud. Depending on the specific requirements, strategies such as 'Rehost', 'Replatform', or 'Refactor' may prove useful.

Choosing the right cloud deployment model: Selecting the right cloud deployment model depends on the unique needs of the business. Today most cloud providers offer three options.

Public cloud: Public clouds are provided by third-party vendors over the internet, such as Amazon AWS, Microsoft Azure, Google Cloud, IBM Cloud, and Digital Ocean. Migrating to the public cloud is well-suited for the 'Rehost' and 'Refactor' strategies.

Private cloud: Private clouds provide cloud services exclusively to a single organisation. These can be on premises, on public cloud infrastructure or through a data centre co-location HCI setup using platforms like VMWare, Nutanix, ProxmoxVE, or OpenStack. For migrating to a private cloud, strategies like 'Replatform' and 'Refactor' may be beneficial.

Hybrid cloud: Hybrid clouds combine public and private cloud models, allowing businesses to leverage the benefits of both. For migrating to a hybrid cloud, strategies such as 'Rehost', 'Replatform', or 'Refactor' can be employed.

Addressing security and compliance concerns: Security and compliance are paramount when moving workloads to the cloud. Here are some migration strategies to address these concerns.

- **Refactor:** Modify the application architecture to take advantage of the security features offered by the cloud environment.
- **Replatform:** Use cloud-native security services and features to enhance security and compliance.
- **Rehost:** Leverage existing security and compliance controls of the application in the cloud environment.

By following the 7Rs of migration, businesses can ensure smooth and secure migration to the cloud environment that meets their specific needs. You can check the AWS Glossary on the 7Rs for ready reference at <https://docs.aws.amazon.com/prescriptive-guidance/latest/migration-retiring-applications/apg-gloss.html#gloss-mig-7rs-terms>.

Data migration

Data migration is a critical aspect of cloud migration -- it is essential to validate the accuracy and integrity of the data during migration. Proper planning and execution, including selecting the appropriate tools and technologies, are important to ensure that data is migrated efficiently and accurately.

Data preparation and secure transfer: Before initiating the transfer process, data needs to be prepared for the cloud environment. This involves organising, cleaning up, and removing duplicates or irrelevant data.

Various methods can be employed for data transfer depending on the data set size.